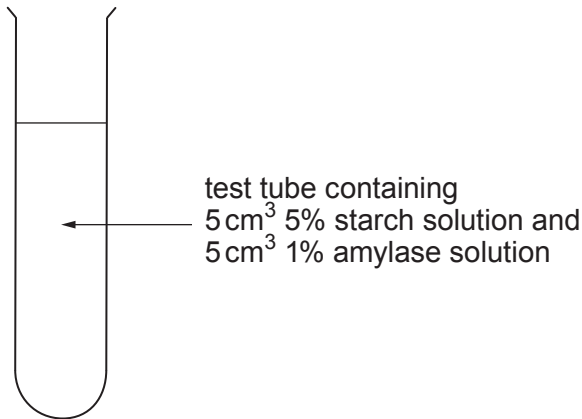


7. Students investigated the activity of amylase enzyme from the bacterium *Bacillus licheniformis* which is used in industry. Amylase digests starch.

They set up six test tubes, each as shown in the diagram below, at a range of temperatures from 10–60°C and measured the amylase activity after 5 minutes.



- (a) The students tried to ensure that their investigation was a fair test.
- State **one other** variable which they should have kept constant in all the test tubes. [1]

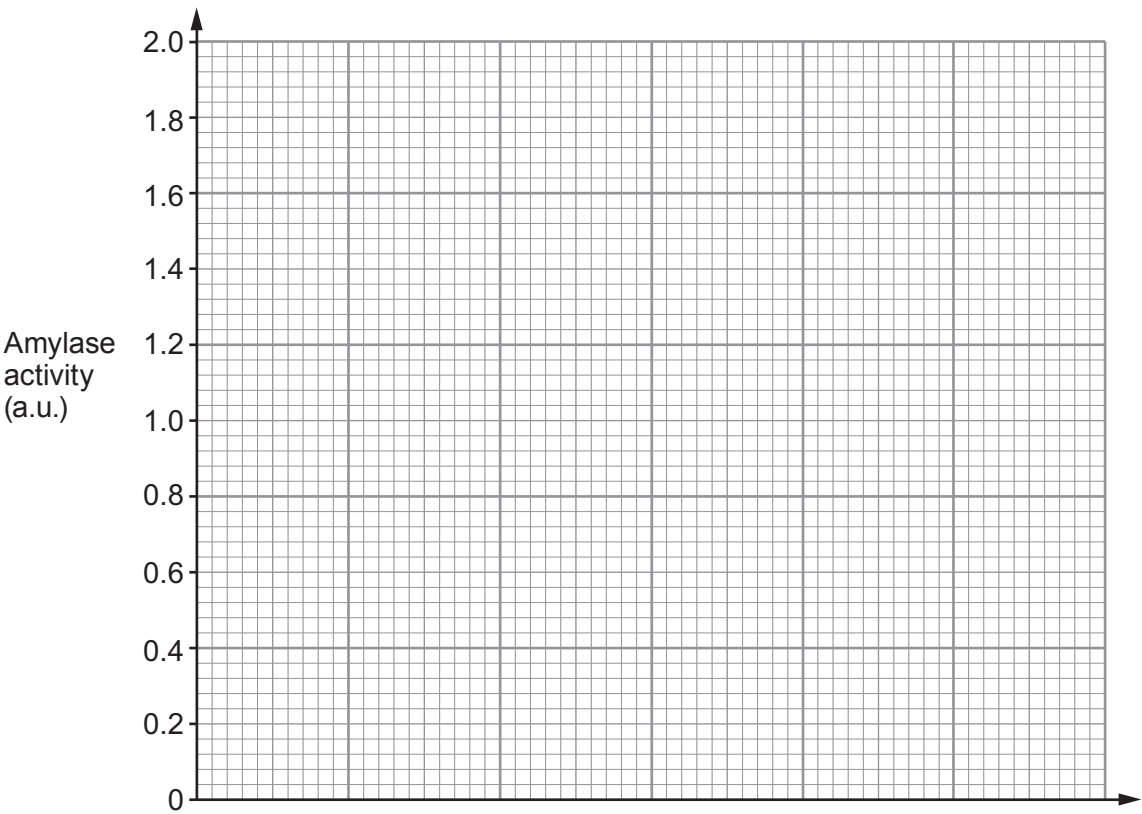
The results of the investigation are shown in the table below.

Temperature (°C)	Amylase activity (a.u.)
10	0.3
20	0.9
30	1.4
35	1.6
40	1.8
60	0.4

- (b) Draw a graph of the results on the grid opposite by
- (i) Choosing a scale for temperature and labelling the axis. [1]
 - (ii) Plotting the points. [2]
 - (iii) Join your plots using a ruler. [1]



Examiner
only



- (c) From your graph,
- (i) State how the amylase activity changes between 15°C and 35°C and explain the reason for this change. [2]
.....
.....
 - (ii) State the reason for the change in the amylase activity between 40°C and 60°C. [1]
.....
 - (iii) I. Use your graph to suggest the optimum temperature for this enzyme. [1]

Optimum temperature = °C

II. State why the optimum temperature cannot be identified with accuracy from these results. Suggest how the investigation could be improved to allow this temperature to be identified more accurately. [2]
.....
.....
.....

